

ABHINAV KUMAR

Contact Information	<i>Email:</i> akumar03@mit.edu <i>Links:</i> Webpage , Google Scholar , Github	
Education	<i>Massachusetts Institute of Technology (MIT)</i> PhD Candidate in Computer Science (EECS) Interests: Causality, Machine Learning Advisor: Prof. Caroline Uhler	2023 - Present GPA: 4.2/5
	<i>Birla Institute Of Technology and Science, Pilani</i> M.Sc.(Hons.) Physics Thesis: Disentangling Mixtures of Unknown Causal Intervention Advisor: Dr. Gaurav Sinha , Adobe Research, Bangalore	2015 - 2020 GPA: 9.54/10
	<i>Birla Institute Of Technology and Science, Pilani</i> B.E.(Hons.) Computer Science Thesis: Fine-Tuning Word Embedding for Domain Adaptation Advisor: Dr. Partha Talukdar , Indian Institute of Science, Bangalore	2016 - 2020 GPA: 9.54/10
Work Experience	<i>Microsoft Research</i> , Bangalore Research Fellow, Advisor: Dr. Amit Sharma Generalization and Explainability of ML model with Causal Perspective	07/21 - 2023
	<i>Adobe Research</i> , Bangalore Research Intern, Advisor: Dr. Gaurav Sinha Analysis of Mixture Models on Causal Graph	01/20 - 07/20
	<i>Indian Institute of Science (IISc)</i> , Bangalore Research Intern, Advisor: Dr. Partha Talukdar Fine-Tuning Word Embedding for Domain Adaptation	05/19 - 12/19
	<i>Google Summer of Code</i> Research Intern, Advisors: Dr. Grasseau Gilles and Dr. Florian Beaudett Deep Learning for Particle Detection and Energy Prediction for particle detectors at CERN	05/18 - 08/18
Publications	1. Abhinav Kumar, Kirankumar Shiragur, Caroline Uhler. “Learning Mixtures of Unknown Causal Interventions”. In 37th Conference on Neural Information Processing Systems (Paper , NeurIPS 2024). 2. Abhinav Kumar, Amit Deshpande, Amit Sharma. “Causal Effect Regularization: Automated Detection and Removal of Spurious Correlations”. 36th Conference on Neural Information Processing Systems (Paper , NeurIPS 2023). 3. Abhinav Kumar, Chenhao Tan, Amit Sharma. “Probing Classifiers are Unreliable for Concept Removal and Detection”. In 35th Conference on Neural Information Processing Systems (Paper , NeurIPS 2022). 4. Abhinav Kumar, Gaurav Sinha. “Disentangling mixtures of unknown causal interventions”. Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence (Paper , UAI 2021) [Oral, 6% acceptance rate].	

5. Aniket Vashishtha, [Abhinav Kumar](#), Atharva Pandey, Abbavaram Gowtham Reddy, Kabir Ahuja, Vineeth N. Balasubramanian, Amit Sharma. “**Teaching Transformers Causal Reasoning through Axiomatic Training**”. Forty-second International Conference on Machine Learning ([Paper ↗](#) , [ICML 2025](#)).
6. Aniket Vashishtha, Abbavaram Gowtham Reddy, [Abhinav Kumar](#), Saketh Bachu, Vineeth N. Balasubramanian, Amit Sharma. “**Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference**”. The Thirteenth International Conference on Learning Representations ([Paper ↗](#) , [ICLR 2024](#)).
7. Gilles Grasseau, [Abhinav Kumar](#), Andrea Sartirana, Artur Lobanov and Florian Beaudette. “**A deep neural network method for analyzing the CMS High Granularity Calorimeter (HGCAL) events**”. 24th International Conference on Computing in High Energy and Nuclear Physics ([Paper ↗](#) , [CHEP 2019](#)).

Relevant Coursework

Computer Science: Statistical Reinforcement Learning (ongoing), Multiagent-Learning (ongoing), Non-Linear Optimization, Fundamentals of Probability, Inference and Information, Algorithm for Inference, Discrete Mathematics, Data Structure and Algorithms, Machine Learning, Data Mining, Information Retrieval, Theory of Computation, Logic in Computer Science, Compilers Construction

Mathematics: Theory of Probability (ongoing), Differential Geometry, Multivariate Calculus, Probability and Statistics, Linear Algebra, Complex Variables and Calculus, Differential Equations

Physics: Statistical Mechanics, Math Methods of Physics, Quantum Mechanics, Non-Linear Dynamics and Chaos, Electromagnetic Theory

Online: [Probabilistic Graphical Models](#), [Deep Learning Specialization](#)

Skills

Programming Languages: Python, C, C++, Java, Matlab, Fortran
Tools and Systems: Tensorflow/Pytorch, Linux, Git